CCNA Networking Concepts

A Study Guide for the CCNA 200-301 Exam

Covers: OSI & TCP/IP Models · IP Addressing & Subnetting · Switching · Routing · WAN Technologies · Security Fundamentals · Wireless · Automation & Programmability

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1. Networking Fundamentals

What Is a Network?

A computer network is a collection of devices connected together to share resources such as files, printers, applications, and internet access. Networks range in size from a small home LAN with a handful of devices to a global service provider backbone connecting millions of endpoints.

Types of Networks

- **LAN (Local Area Network):** Covers a small geographic area such as a home, office, or building. Typically owned and managed by a single organization.
- **WAN (Wide Area Network):** Connects LANs across large geographic distances, often using a service provider's infrastructure.
- **MAN (Metropolitan Area Network):** Spans a city or large campus, larger than a LAN but smaller than a WAN.
- **WLAN (Wireless LAN):** A LAN that uses radio frequency (Wi-Fi) instead of cabling to connect devices.
- **PAN (Personal Area Network):** A very short-range network such as Bluetooth between a phone and headset.
- **SAN (Storage Area Network):** A dedicated high-speed network providing access to consolidated block-level storage.

Network Topologies

- **Bus:** All devices share a single cable. Rarely used today; a single break disrupts the entire segment.
- **Star:** All devices connect to a central switch or hub. The most common physical topology in modern LANs.
- **Ring:** Each device connects to exactly two others, forming a loop. Historically used in Token Ring/FDDI.
- **Mesh (full/partial):** Devices have multiple interconnections for redundancy, common in WAN and data center core designs.
- **Hybrid:** A combination of topologies, such as a star of stars, common in hierarchical campus designs.

Core Network Devices

Device	Layer	Primary Function
Hub	Physical (L1)	Repeats electrical signal to all ports; single collision domain.
Switch	Data Link (L2)	Forwards frames based on MAC address; each port is its own collision domain.
Router	Network (L3)	Forwards packets between different networks based on IP address.
Firewall	L3/L4 (and higher)	Enforces security policy by filtering traffic between zones.
Access Point	Physical/Data Link	Bridges wireless clients onto the wired LAN.
Layer 3 Switch	L2 and L3	Combines high-speed switching with inter-VLAN routing.

Collision Domains vs. Broadcast Domains

A **collision domain** is a segment where frames can collide if two devices transmit at the same time; each switch port is its own collision domain, effectively eliminating collisions on full-duplex links. A **broadcast domain** is the set of devices that receive a broadcast frame sent by any member; routers (and VLANs) separate broadcast domains, while switches by default do not.

Duplex and Speed

Half-duplex allows a device to send or receive but not both simultaneously, and can cause collisions. Full-duplex allows simultaneous send and receive and is standard on modern switched links, eliminating collisions on that link. Duplex mismatches (one side full, one side half) are a classic cause of intermittent performance problems and should always be checked when troubleshooting slow links.

Ethernet Cabling and Media

Cable/Media	Max Distance	Typical Speed	Notes
Cat5e (UTP)	100 m	1 Gbps	Common baseline for gigabit Ethernet.
Cat6 (UTP)	100 m (55 m for 10G)	1-10 Gbps	Better shielding/twist rate than Cat5e.
Cat6a (UTP)	100 m	10 Gbps	Full 10 Gbps over 100 m.
Multimode fiber	Up to ~550 m (varies)	1-100+ Gbps	Uses LED/laser; shorter reach, lower cost than single-m
Single-mode fiber	Kilometers	1-100+ Gbps	Longer reach, used for WAN and long campus backbo

Twisted-pair copper cabling uses twisted wire pairs to reduce electromagnetic interference (EMI) and crosstalk between pairs. Fiber optic cabling is immune to EMI entirely since it transmits light rather than electrical signals, making it well suited to noisy environments and long distances.

2. The OSI and TCP/IP Models

Why Reference Models Matter

Reference models break networking into layers so that engineers can isolate problems, and vendors can build interoperable products. The OSI model is a conceptual, seven-layer teaching model; the TCP/IP model is the practical four-layer model that actually describes how the modern internet works.

The OSI Model (7 Layers)

#	Layer	Unit (PDU)	Examples / Notes
7	Application	Data	HTTP, DNS, DHCP, SMTP — end-user facing protocols.
6	Presentation	Data	Encryption, compression, character encoding (e.g., TLS, ASCII/JPEG).
5	Session	Data	Establishes, manages, and tears down sessions (e.g., NetBIOS, RPC).
4	Transport	Segment	TCP (reliable) and UDP (best-effort); ports; flow control.
3	Network	Packet	IP addressing and routing; ICMP.
2	Data Link	Frame	MAC addressing, switching; Ethernet, ARP resolution happens here (L2/L3 boundary).
1	Physical	Bits	Cabling, connectors, voltages, radio signals.

Mnemonic:

Top-down: *All People Seem To Need Data Processing*. Bottom-up: *Please Do Not Throw Sausage Pizza Away*.

The TCP/IP Model (4 Layers)

TCP/IP Layer	Maps to OSI	Key Protocols
Application	App, Presentation, Session	HTTP/HTTPS, DNS, DHCP, FTP, SSH, SMTP
Transport	Transport	TCP, UDP
Internet	Network	IP, ICMP, ARP
Network Access	Data Link, Physical	Ethernet, Wi-Fi (802.11)

Encapsulation and De-encapsulation

As data moves down the stack on the sending device, each layer adds its own header (and sometimes a trailer), a process called **encapsulation**: Data becomes a Segment (L4), then a Packet (L3), then a Frame (L2), then Bits (L1). The receiving device reverses this process, called **de-encapsulation**, stripping headers as data moves back up the stack.

TCP vs. UDP

Characteristic	TCP	UDP
Connection	Connection-oriented (3-way handshake)	Connectionless

Characteristic	TCP	UDP
Reliability	Reliable — acknowledgments, retransmission	Best-effort, no guarantee of delivery
Ordering	Guarantees in-order delivery	No ordering guarantee
Overhead	Higher (headers, sequencing, flow control)	Lower, minimal header
Typical Use	Web (HTTP), email, file transfer	DNS, DHCP, VoIP, video streaming

The TCP Three-Way Handshake

- **SYN:** Client sends a segment with the SYN flag set to request a connection.
- **SYN-ACK:** Server responds with SYN and ACK flags set, acknowledging the request and proposing its own sequence number.
- **ACK:** Client acknowledges the server's SYN, and the connection is now established.

3. IP Addressing and Subnetting

IPv4 Address Structure

An IPv4 address is a 32-bit number written as four decimal octets (0-255) separated by dots, e.g., 192.168.1.10. Each address has a **network portion** and a **host portion**, determined by the subnet mask.

IPv4 Address Classes (Legacy Reference)

Class	First Octet Range	Default Mask	Typical Use
A	1 - 126	255.0.0.0 (/8)	Very large networks
B	128 - 191	255.255.0.0 (/16)	Medium networks
C	192 - 223	255.255.255.0 (/24)	Small networks
D	224 - 239	N/A	Multicast
E	240 - 255	N/A	Experimental/Reserved

Private (RFC 1918) Address Ranges

Range	CIDR	Class
10.0.0.0 - 10.255.255.255	10.0.0.0/8	A
172.16.0.0 - 172.31.255.255	172.16.0.0/12	B
192.168.0.0 - 192.168.255.255	192.168.0.0/16	C

CIDR Notation and Subnet Masks

CIDR (Classless Inter-Domain Routing) notation expresses the mask as a slash followed by the number of network bits, e.g., /24 means the first 24 bits are network bits (255.255.255.0). This is the modern way subnet masks are written, replacing rigid class boundaries.

Subnetting Cheat Sheet (Class C / small blocks)

CIDR	Subnet Mask	# Subnets (from /24)	Hosts per Subnet
/24	255.255.255.0	1	254
/25	255.255.255.128	2	126
/26	255.255.255.192	4	62
/27	255.255.255.224	8	30
/28	255.255.255.240	16	14
/29	255.255.255.248	32	6
/30	255.255.255.252	64	2
/31	255.255.255.254	128	2 (point-to-point, RFC 3021)

CIDR	Subnet Mask	# Subnets (from /24)	Hosts per Subnet
/32	255.255.255.255	256	1 (host route)

Subnetting Method — Quick Steps

- 1. Identify the number of host bits needed (2^n – 2 ≥ required hosts, for usable hosts).
- 2. Determine the new mask by borrowing bits from the host portion.
- 3. Calculate the **block size** = 256 – (interesting octet value of the mask).
- 4. List subnet boundaries by adding the block size repeatedly to find network, broadcast, and usable ranges.
- 5. Example: 192.168.10.0/26 → block size 64 → subnets are .0, .64, .128, .192 (each with 62 usable hosts).

VLSM (Variable Length Subnet Masking)

VLSM allows different subnets within the same network to use different mask lengths, so address space is allocated efficiently based on actual host requirements rather than a single fixed mask for the whole network. Best practice is to allocate the largest subnets first, then work down to the smallest (e.g., point-to-point /30 or /31 links).

IPv6 Fundamentals

- 128-bit address written in eight groups of four hex digits, separated by colons, e.g., 2001:0db8:0000:0000:0000:ff00:0042:8329.
- Leading zeros in a group may be omitted, and one contiguous run of all-zero groups may be replaced with '::' (only once per address).
- **Unicast:** one-to-one; **Multicast:** one-to-many; **Anycast:** one-to-nearest. IPv6 has no broadcast.
- **Link-local** addresses (fe80::/10) are automatically assigned and used for on-link communication such as neighbor discovery.
- **Global unicast** addresses (2000::/3) are the IPv6 equivalent of public IPv4 addresses, routable on the internet.
- SLAAC (Stateless Address Autoconfiguration) lets a host self-assign an address using router advertisements, without DHCP.

4. Switching Concepts

How Switches Forward Frames

A switch builds and maintains a **MAC address table** (CAM table) by inspecting the source MAC address of every frame it receives and associating it with the incoming port. When forwarding, the switch looks up the destination MAC: if it's known, the frame is sent only out that port; if unknown, the frame is flooded out all ports except the one it arrived on.

VLANs (Virtual LANs)

A VLAN logically segments a single physical switch (or group of switches) into multiple separate broadcast domains. Devices in different VLANs cannot communicate without a Layer 3 device (router or L3 switch) to route between them. Common uses include separating voice, data, guest, and management traffic.

- **Access port:** Carries traffic for a single VLAN; connects to end devices such as PCs and printers.
- **Trunk port:** Carries traffic for multiple VLANs between switches, tagging frames with 802.1Q headers.
- **Native VLAN:** The one VLAN on a trunk that is sent untagged (default VLAN 1 unless changed).
- **Voice VLAN:** A separate VLAN for IP phones, allowing data and voice to share one access port safely.

802.1Q Trunking

802.1Q inserts a 4-byte tag into the Ethernet frame identifying the VLAN ID (1-4094), allowing a single physical trunk link to carry traffic for many VLANs. Frames belonging to the native VLAN are typically left untagged.

Spanning Tree Protocol (STP)

STP (IEEE 802.1D) prevents Layer 2 loops in networks with redundant switch links, which would otherwise cause broadcast storms and MAC table instability. STP elects a **root bridge** (lowest bridge ID) and calculates the lowest-cost path from every switch to that root, blocking redundant paths to prevent loops while keeping them available as backups.

Port State	Function
Blocking	Does not forward frames; listens for BPDUs only.
Listening	Transitioning state; still no frame forwarding, prepares to become root/designated.
Learning	Builds the MAC address table but still does not forward frames.
Forwarding	Fully operational — sends and receives data frames.
Disabled	Administratively shut down.

Rapid PVST+ (802.1w) is the modern, faster-converging variant widely deployed today, using port roles such as Root, Designated, Alternate, and Backup instead of the slower legacy states.

EtherChannel / Link Aggregation

EtherChannel bundles multiple physical links between two switches into one logical link, increasing bandwidth and providing redundancy without STP blocking any of the bundled ports. It can be negotiated

dynamically with LACP (open standard, IEEE 802.3ad) or PAgP (Cisco proprietary), or configured statically ('on').

Port Security

Port security restricts which MAC addresses may connect to a switch port, limiting the number of allowed MACs and defining a violation action (protect, restrict, or shutdown/err-disable) if that limit is exceeded — a basic defense against rogue devices and MAC flooding attacks.

5. Routing Concepts

How Routers Forward Packets

A router examines the destination IP address of each packet and consults its **routing table** to find the best matching route (longest prefix match wins), then forwards the packet out the associated interface toward the next hop. If no matching route exists and there is no default route, the packet is dropped.

Administrative Distance

When multiple sources offer a route to the same destination, the router prefers the route with the lowest **administrative distance** (a measure of trustworthiness of the source), regardless of metric.

Route Source	Default Administrative Distance
Directly connected	0
Static route	1
EIGRP (internal)	90
OSPF	110
IS-IS	115
RIP	120
EIGRP (external)	170
Unknown / unreachable	255 (never used)

Static vs. Dynamic Routing

- **Static routing:** Manually configured routes. Predictable and secure but doesn't adapt to topology changes and doesn't scale well.
- **Dynamic routing:** Routing protocols automatically learn and share routes, adapting to link failures and topology changes.

Interior Gateway Protocol Comparison

Protocol	Type	Metric	Notes
RIP(v2)	Distance-vector	Hop count (max 15)	Simple but limited; largely legacy today.
EIGRP	Advanced distance-vector	Bandwidth/delay composite	Cisco-originated; fast convergence via DUAL algorithm.
OSPF	Link-state	Cost (based on bandwidth)	Open standard; widely used in enterprise networks; use
IS-IS	Link-state	Cost	Common in large service-provider networks.
BGP	Path-vector	Path attributes	The routing protocol of the internet; used between autor

OSPF Basics

- Routers form neighbor adjacencies and exchange Link-State Advertisements (LSAs) to build an identical topology database (LSDB).
- The Shortest Path First (SPF/Dijkstra) algorithm calculates the best path from that database.
- OSPF uses **areas** to limit LSA flooding scope; Area 0 (the backbone) must exist and all other areas typically connect to it.
- Router IDs are chosen from the highest loopback IP, or highest active IP if no loopback is configured.
- Default reference bandwidth is 100 Mbps for cost calculation, often adjusted upward on modern high-speed networks.

Default Route and Gateway of Last Resort

A default route (0.0.0.0/0 in IPv4) matches any destination not covered by a more specific route, and is commonly used to point traffic toward the internet edge or upstream provider.

First Hop Redundancy Protocols (FHRP)

FHRPs such as HSRP (Cisco), VRRP (open standard), and GLBP provide a shared virtual gateway IP across two or more routers, so that if the active router fails, a standby router seamlessly takes over without hosts needing to change their configured default gateway.

6. WAN Technologies and Virtualization

WAN Connection Types

- **Leased line:** Dedicated point-to-point circuit from a provider, always available, but relatively costly.
- **MPLS:** Provider-based label-switching service that can carry multiple sites' traffic across a shared backbone with defined paths.
- **Internet VPN:** Uses the public internet with encryption (e.g., IPsec) to create secure site-to-site or remote-access tunnels, at lower cost than private circuits.
- **Metro Ethernet:** Ethernet-based WAN service delivered by a provider within a metropolitan area.
- **DSL/Cable/Fiber broadband:** Common for branch or remote-access connectivity, often paired with a VPN overlay.

VPNs

- **Site-to-site VPN:** Connects two networks (e.g., branch to HQ) over an untrusted network, typically using IPsec.
- **Remote-access VPN:** Connects an individual client device to a corporate network, often via SSL/TLS or IPsec client software.
- **GRE tunnel:** Encapsulates a wide variety of protocols for transport but by itself provides no encryption (often paired with IPsec).

Virtualization Concepts

- **Virtual machine (VM):** A software-based computer running on a hypervisor, sharing physical host hardware.
- **Hypervisor:** Type 1 (bare-metal, runs directly on hardware) vs. Type 2 (hosted, runs atop an OS).
- **Virtual switch (vSwitch):** A software switch inside a hypervisor connecting VMs to each other and to physical NICs.
- **Container:** A lightweight, OS-level virtualization unit that packages an application with its dependencies, without a full guest OS per instance.

Cloud Deployment and Service Models

Model	What the Provider Manages	Example Use
IaaS	Physical infrastructure, virtualization	Rent virtual servers/storage/networking
PaaS	Infrastructure + runtime/platform	Deploy code without managing servers
SaaS	Everything, including the application	Use a hosted email or CRM application

SD-WAN and Controller-Based Networking

Software-Defined WAN (SD-WAN) centralizes policy and path-selection decisions in a controller, abstracting the underlying transport (MPLS, broadband, LTE) and dynamically choosing the best path per application. This reflects the broader industry shift toward controller-based (SDN) architectures, which separate the **control plane** (decision making) from the **data plane** (actual packet forwarding).

7. Security Fundamentals

Core Security Concepts

- **Confidentiality, Integrity, Availability (CIA triad):** The foundational goals of information security.
- **Authentication, Authorization, Accounting (AAA):** Verifying identity, granting permissions, and logging activity, often via RADIUS or TACACS+.
- **Defense in depth:** Layering multiple security controls so that no single failure compromises the whole network.

Access Control Lists (ACLs)

An ACL is an ordered list of permit/deny statements that filters traffic based on criteria such as source/destination IP address, protocol, and port number. Rules are processed top-down, and processing stops at the first match; an implicit deny exists at the end of every ACL.

- **Standard ACL:** Filters based on source IP address only; typically placed close to the destination.
- **Extended ACL:** Filters based on source/destination IP, protocol, and port; typically placed close to the source.
- **Named ACL:** Uses a descriptive name instead of a number, and allows easier editing of individual entries.

Common Threats and Mitigations

Threat	Description	Common Mitigation
MAC flooding	Overwhelms the CAM table to force flooding, enabling sniffing.	Port security (MAC limits)
DHCP spoofing	Rogue DHCP server hands out malicious configuration.	DHCP snooping
ARP spoofing	Attacker poisons ARP caches to intercept traffic.	Dynamic ARP Inspection (DAI)
VLAN hopping	Attacker crafts frames to jump between VLANs.	Disable auto-trunking, set unused native VLAN
Denial of Service	Floods a target with traffic/requests to exhaust resources.	Rate limiting, firewalls, IPS
Phishing/social engineering	Tricks users into revealing credentials or installing malware.	User training, email filtering, MFA

Firewalls and Segmentation

Firewalls enforce policy at trust boundaries, ranging from simple stateless packet filters to stateful firewalls that track connection state, and next-generation firewalls that add application awareness and intrusion prevention. Network segmentation (via VLANs, subnets, and zones such as a DMZ) limits how far an attacker can move if one segment is compromised.

Device Hardening Basics

- Change default credentials and disable unused services/ports.
- Use SSH instead of Telnet for management access (Telnet sends credentials in clear text).
- Apply the principle of least privilege for administrative accounts.
- Keep software/firmware patched and maintain configuration backups.

- Use dedicated management VLANs/networks separate from user traffic.

8. Wireless Networking

802.11 Standards Overview

Standard	Wi-Fi Name	Band	Max Theoretical Speed
802.11a	-	5 GHz	54 Mbps
802.11b	-	2.4 GHz	11 Mbps
802.11g	-	2.4 GHz	54 Mbps
802.11n	Wi-Fi 4	2.4/5 GHz	600 Mbps
802.11ac	Wi-Fi 5	5 GHz	~3.5 Gbps
802.11ax	Wi-Fi 6/6E	2.4/5/6 GHz	~9.6 Gbps

Wireless Architecture

- **Autonomous AP:** Standalone access point managing its own configuration and radio.
- **Lightweight AP + WLC:** APs forward control traffic to a central Wireless LAN Controller using CAPWAP, simplifying management at scale.
- **SSID:** The broadcast network name that clients see and select to associate with a WLAN.
- **BSS/ESS:** A Basic Service Set is one AP's coverage cell; an Extended Service Set links multiple APs sharing an SSID for roaming.

Wireless Security

Security Type	Notes
Open	No encryption; traffic sent in the clear.
WEP	Legacy, cryptographically broken — should not be used.
WPA2-Personal (PSK)	Pre-shared key with AES-CCMP encryption; common for home/small office.
WPA2/WPA3-Enterprise	Uses 802.1X with a RADIUS server for per-user authentication.
WPA3	Current standard; adds stronger handshake (SAE) resistant to offline dictionary attacks.

RF Concepts

- **Channel:** A specific frequency slice within a band; adjacent APs should use non-overlapping channels to reduce interference (e.g., 1, 6, 11 on 2.4 GHz).
- **Co-channel interference:** Occurs when multiple APs share the same channel and contend for airtime.
- **Signal-to-noise ratio (SNR):** The strength of the desired signal relative to background noise; higher is better.

9. Network Services: DHCP, DNS, and NAT

DHCP (Dynamic Host Configuration Protocol)

DHCP automatically assigns IP addressing information to hosts, removing the need for manual configuration. A DHCP server hands out an address along with related settings such as subnet mask, default gateway, DNS servers, and lease duration.

The DORA Process

- **Discover:** Client broadcasts a request looking for any DHCP server.
- **Offer:** A DHCP server responds with a proposed IP address and configuration.
- **Request:** The client broadcasts a request confirming it wants to accept a specific offer.
- **Acknowledge:** The server confirms the lease with a DHCPACK, finalizing the assignment.

A **DHCP relay agent** (ip helper-address on Cisco IOS) forwards DHCP broadcasts across routed boundaries to a centralized DHCP server, since routers do not forward broadcast traffic by default.

DNS (Domain Name System)

DNS translates human-friendly domain names (e.g., example.com) into IP addresses that devices use to actually route traffic. It is a distributed, hierarchical database.

- **Root servers:** Top of the hierarchy; direct queries to the appropriate top-level domain (TLD) servers.
- **Authoritative server:** Holds the actual DNS records for a specific domain.
- **Recursive resolver:** Queries on behalf of the client, walking the hierarchy until it finds the answer.
- **A record:** Maps a hostname to an IPv4 address. **AAAA record:** Maps a hostname to an IPv6 address.
- **CNAME record:** An alias pointing one hostname to another hostname.
- **MX record:** Specifies the mail server responsible for a domain.
- **PTR record:** Used for reverse lookups, mapping an IP address back to a hostname.

NAT (Network Address Translation)

NAT translates private, internal IP addresses to a public address (or a different address) as traffic crosses a boundary, most commonly allowing many internal hosts to share a small number of public addresses when reaching the internet.

NAT Type	Description
Static NAT	One-to-one, permanent mapping between a private and a public address.
Dynamic NAT	Maps private addresses to a pool of public addresses on a first-come basis.
PAT (NAT Overload)	Many-to-one; uses port numbers to let many internal hosts share one public IP — the most common form i

- **Inside local:** A private address as seen inside the network.
- **Inside global:** The translated, public-facing address as seen outside the network.

10. Network Automation and Programmability

Why Automation Matters

Traditional device-by-device CLI configuration doesn't scale well and is prone to human error. Automation applies consistent, version-controlled configuration across many devices quickly and predictably, which is an increasingly emphasized topic on the modern CCNA exam.

Control Plane vs. Data Plane vs. Management Plane

- **Data plane:** Actually forwards user traffic (frames/packets) through the device.
- **Control plane:** Makes forwarding decisions — routing protocols, STP, ARP.
- **Management plane:** Used to configure and monitor the device itself — SSH, SNMP, NETCONF.

Software-Defined Networking (SDN)

SDN centralizes the control plane in a controller that has a broad view of the network and pushes forwarding instructions to devices, decoupling network intelligence from individual hardware boxes. This enables more centralized policy, faster changes, and programmatic management compared to purely device-by-device control.

APIs and Data Formats

- **REST API:** The most common way modern network controllers and devices expose programmatic access, typically using HTTP methods (GET, POST, PUT, DELETE) against defined resources.
- **JSON:** A lightweight, human-readable data-interchange format widely used in REST API request/response bodies.
- **YAML:** A human-friendly data format commonly used for automation playbooks and configuration files.
- **XML:** An older markup-based data format still used by some network management protocols such as NETCONF.

Configuration Management Tools

Tool	Model	Notes
Ansible	Agentless, push-based	Uses SSH; configuration defined in YAML playbooks.
Puppet	Agent-based, pull-based	Agents periodically pull configuration from a master.
Chef	Agent-based, pull-based	Configuration defined as Ruby-based 'recipes'.
Terraform	Agentless, declarative	Widely used for infrastructure-as-code provisioning.

11. Quality of Service (QoS) Basics

Why QoS Matters

QoS mechanisms manage bandwidth, delay, jitter, and loss so that important or sensitive traffic (like voice and video) is treated preferentially over less time-sensitive traffic (like bulk file transfers), especially on links where congestion can occur.

Key Traffic Characteristics

- **Bandwidth:** The capacity of a link, measured in bits per second.
- **Delay (latency):** Time for a packet to travel from source to destination.
- **Jitter:** Variation in delay between packets — especially disruptive for real-time voice/video.
- **Loss:** Packets that fail to arrive, often due to congestion or errors.

QoS Models

Model	Description
Best Effort	No QoS applied; all traffic treated equally (the internet default).
IntServ	Reserves guaranteed resources end-to-end per flow (e.g., RSVP); doesn't scale well.
DiffServ	Classifies and marks traffic into a small number of classes for per-hop treatment; the most widely deployed mod

Common QoS Mechanisms

- **Classification and marking:** Identifying traffic type and tagging it (e.g., DSCP values in the IP header, CoS in the 802.1Q tag).
- **Queuing:** Determines the order packets leave an interface during congestion (e.g., Low Latency Queuing prioritizes voice).
- **Policing:** Drops or re-marks traffic that exceeds a defined rate, typically applied at ingress.
- **Shaping:** Buffers and smooths excess traffic to conform to a rate rather than dropping it outright.
- **Congestion avoidance:** Proactively drops some packets before a queue fully fills (e.g., WRED) to prevent global synchronization of TCP flows.

12. Troubleshooting Tools and Methodology

Systematic Troubleshooting Approach

- 1. **Define the problem** — gather symptoms and identify what changed.
- 2. **Establish a theory of probable cause** — form a hypothesis.
- 3. **Test the theory** — confirm or rule it out.
- 4. **Establish an action plan** and implement the fix.
- 5. **Verify full functionality** and, if applicable, implement preventive measures.
- 6. **Document** findings, actions, and outcomes.

OSI-Based Troubleshooting Approaches

- **Bottom-up:** Start at Layer 1 (cabling, link lights) and work upward. Good when the cause is unclear.
- **Top-down:** Start at Layer 7 (the application) and work downward. Good when the problem is likely application-specific.
- **Divide-and-conquer:** Start in the middle (commonly Layer 3) and move up or down based on results. Efficient when you have a hunch.

Cisco IOS CLI Modes

Mode	Prompt Example	Purpose
User EXEC	Router>	Limited, view-only mode; basic monitoring commands.
Privileged EXEC	Router#	Full monitoring access; entered via 'enable'.
Global Configuration	Router(config)#	Device-wide settings; entered via 'configure terminal'.
Interface Configuration	Router(config-if)#	Settings specific to one interface.
Line Configuration	Router(config-line)#	Settings for console/VTY/AUX lines (e.g., passwords, timeouts).

Configuration changes made in these modes affect the **running-config** (active memory) immediately. Use 'copy running-config startup-config' (or 'write memory') to save changes so they survive a reload.

Common CLI Diagnostic Commands

Command	Purpose
ping	Tests basic reachability using ICMP echo request/reply.
traceroute / tracert	Shows the hop-by-hop path packets take to a destination.
ipconfig / ifconfig / ip addr	Displays local host IP configuration.
show ip interface brief	Quick summary of interface status and IP addresses (Cisco IOS).
show mac address-table	Displays the switch's learned MAC-to-port mappings (Cisco IOS).
show ip route	Displays the router's routing table (Cisco IOS).
nslookup / dig	Queries DNS to resolve hostnames or troubleshoot DNS issues.

Command	Purpose
show interfaces	Displays interface counters, errors, and status (Cisco IOS).

Cable and Physical Layer Checks

- Verify link lights and check for excessive interface errors (CRC errors, collisions) with show interfaces.
- Confirm correct cable type — straight-through for dissimilar devices, crossover for similar devices (though most modern NICs auto-detect via Auto-MDIX).
- Check for duplex/speed mismatches between connected devices.

Sample Troubleshooting Scenario

Symptom: A user reports they can reach devices on their local subnet but cannot reach anything outside it.

- **Check 1 — Layer 1/2:** Confirm the link light is up and the switch port isn't err-disabled.
- **Check 2 — Layer 3 local:** Verify the host's IP, subnet mask, and default gateway are correctly configured.
- **Check 3 — Gateway reachability:** Ping the default gateway directly; if it fails, the problem is likely local (wrong gateway, VLAN mismatch, or ARP issue).
- **Check 4 — Upstream routing:** If the gateway responds but nothing beyond it does, check the router's routing table for a route to the destination or a default route.
- **Check 5 — DNS vs. connectivity:** If pinging by IP works but a URL doesn't resolve, the issue is DNS, not IP connectivity — check nslookup/dig output.

13. Quick Reference Cheat Sheets

Common Well-Known Ports

Port	Protocol	Service
20/21	TCP	FTP (data/control)
22	TCP	SSH
23	TCP	Telnet
25	TCP	SMTP
53	TCP/UDP	DNS
67/68	UDP	DHCP (server/client)
69	UDP	TFTP
80	TCP	HTTP
110	TCP	POP3
123	UDP	NTP
143	TCP	IMAP
161/162	UDP	SNMP (agent/trap)
443	TCP	HTTPS
514	UDP	Syslog
3389	TCP	RDP

Binary-to-Decimal Octet Reference

Powers of two used when converting subnet masks and calculating host ranges: 128, 64, 32, 16, 8, 4, 2, 1 (these sum to 255 across a full octet, and are the basis for all subnet mask math).

Mask-to-CIDR Quick Table

Octet Value	Binary	CIDR bits contributed
255	11111111	8
254	11111110	7
252	11111100	6
248	11111000	5
240	11110000	4
224	11100000	3
192	11000000	2

Octet Value	Binary	CIDR bits contributed
128	10000000	1
0	00000000	0

Key Exam-Ready Reminders

- The subnet's **network address** is always the first address in the range (all host bits = 0); the **broadcast address** is the last (all host bits = 1). Neither is normally assignable to a host.
- Longest prefix match always wins in the routing table, regardless of administrative distance or metric.
- A switch never routes between VLANs by itself — inter-VLAN routing requires a router or Layer 3 switch (a 'router on a stick' or SVI configuration).
- STP blocks ports to prevent loops; it does not disable the physical link, only its forwarding of data frames.
- Private IP ranges (RFC 1918) are not routable on the public internet and require NAT/PAT to reach it.

14. Practice Review Questions

Test your understanding of the concepts above. Answers are listed at the end of this section.

- 1. Which OSI layer is responsible for logical addressing and routing between networks?
- 2. What is the usable host range for the subnet 192.168.5.64/27?
- 3. Which protocol prevents Layer 2 loops in a switched network with redundant links?
- 4. What administrative distance does a directly connected route have by default?
- 5. Which TCP/IP model layer corresponds to the OSI Session, Presentation, and Application layers combined?
- 6. What DHCP message does a client broadcast to accept a specific server's offer?
- 7. Which NAT type allows many internal hosts to share a single public IP address using port numbers?
- 8. What is the purpose of the native VLAN on an 802.1Q trunk?
- 9. Which wireless security standard replaced WPA2 and uses SAE for a stronger handshake?
- 10. In routing, which wins when two routes to the same destination have different prefix lengths?

Answer Key

- 1. Network layer (Layer 3).
- 2. 192.168.5.65 through 192.168.5.94 (block size 32; network .64, broadcast .95).
- 3. Spanning Tree Protocol (STP / Rapid PVST+ / 802.1w).
- 4. 0 (most trusted route source).
- 5. Application layer (of the TCP/IP model).
- 6. DHCPREQUEST (the 'Request' step of DORA).
- 7. PAT (Port Address Translation / NAT overload).
- 8. It carries untagged traffic across the trunk (the default VLAN for frames without an 802.1Q tag).
- 9. WPA3.
- 10. The route with the longest prefix match (most specific) always wins, regardless of administrative distance.

Study Tips for Exam Success

- **Build a home lab or use a simulator:** Hands-on practice with real or simulated CLI (e.g., Packet Tracer, GNS3) cements concepts far better than reading alone.
- **Practice subnetting until it's automatic:** Timed subnetting drills are one of the highest-value ways to prepare, since address math questions appear throughout the exam.
- **Draw topologies:** Sketching network diagrams while studying routing and switching helps visualize traffic flow and troubleshoot logically.
- **Study in layers:** Revisit the OSI model as an organizing framework — tie every new topic back to the layer it operates at.
- **Use spaced repetition:** Revisit earlier chapters periodically rather than only moving forward, to keep foundational concepts fresh.
- **Take practice exams under time pressure:** This builds pacing skills and reveals weak areas before exam day.
- **Read the current official exam topics list:** Cisco periodically updates the CCNA blueprint, so always confirm coverage against the latest version before your exam date.

End of study guide. These notes are a study aid and do not replace official Cisco curriculum, hands-on lab practice, and the official exam topics list, which should always be checked for the most current version of the exam blueprint.